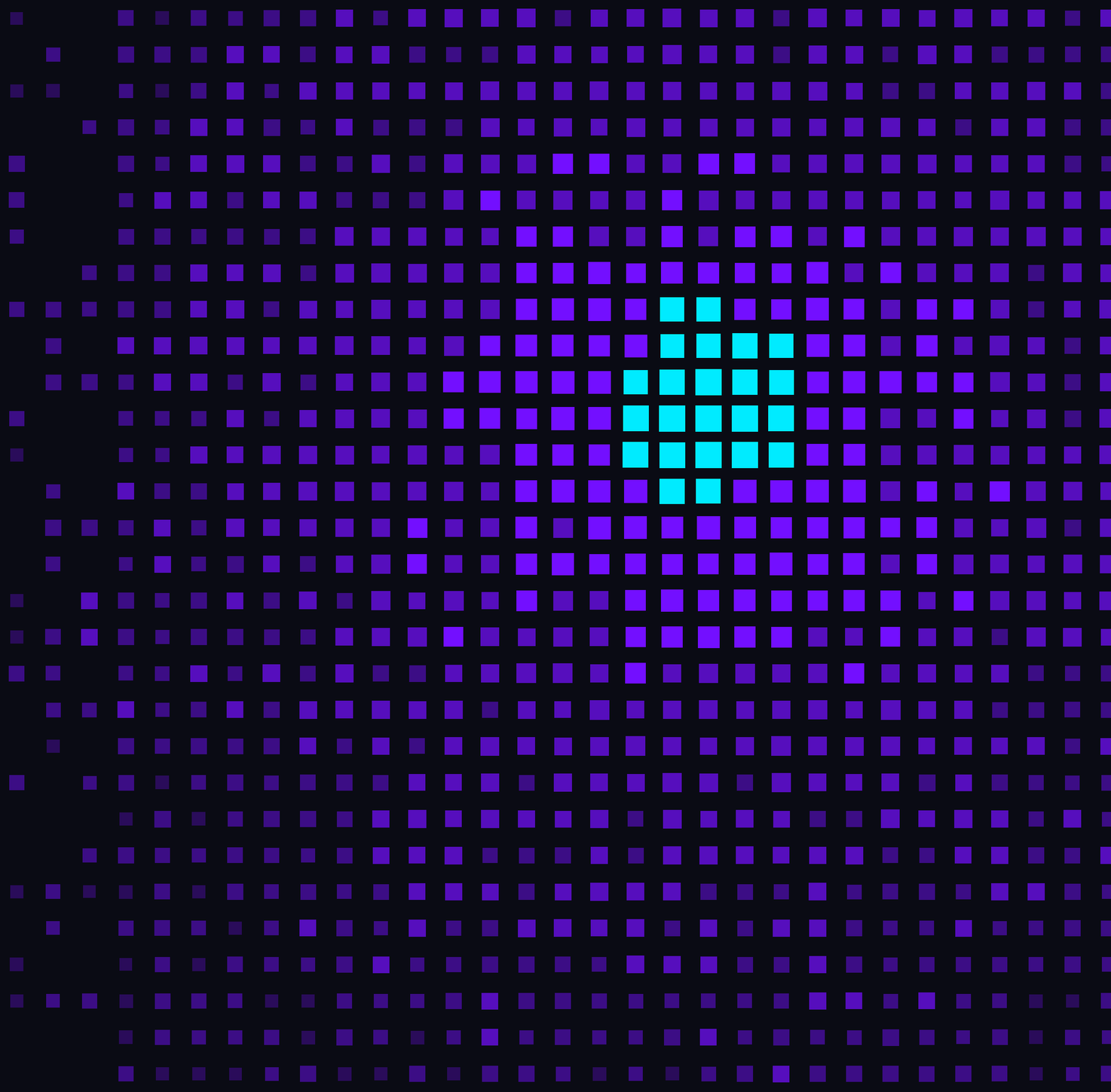
ATRC ADVANCED TECHNOLOGY PIONEERS 2026 · EDGE TRACK

INFINET

إنفينت

Stay connected. Stay secure.

Infinet keeps your call alive as you move between Wi-Fi, mobile, satellite and wired networks, and never hands it to a network it hasn't checked.



TEAM INFINET · 42 ABU DHABI

NSTI FESTIVAL · ADNEC AL AIN · STAND 31

TODAY'S PHONE · SAME WALK, SAME SIGNAL

Moving between networks still freezes calls, and phones trust any network with a familiar name.

Leaving a building

6.56s

frozen on average, waiting for the Wi-Fi to die

202 / 300 calls survive

A fake café Wi-Fi

11.45s

frozen on average, joining a copy of the café's network

882 voice frames a call sent readable to the copy

A desert site on satellite

12.68s

frozen on average when the satellite link fails over

60 / 300 calls survive

THE CHALLENGE

How might we keep a user's experience unbroken as they move between Wi-Fi, mobile, satellite and wired networks? EDGE challenge statement

SIMULATED · 300 RUNS PER SCENE · TODAY'S PHONE SWITCHES AFTER THE DROP

THREE FEATURES

See the drop coming. Check before trusting. Keep the call.

01 · PREDICT

Sees the drop coming

Our coverage map knows where the Wi-Fi ends. Heading for that edge gives seconds of warning, not fractions of one.

3.4 s early, on average

02 · TRUST

Checks before it trusts

Every new network passes a check before our traffic touches it. A fake network never gets the call.

0 frames over a fake network in 5,000 runs

03 · TUNNEL

Keeps the call

One encrypted tunnel to our anchor server, so changing networks never breaks the session.

900 / 900 calls kept across three scenes

Prediction buys time. Infinet spends it on trust.

SYSTEM DESIGN

Every network only ever carries sealed packets to our anchor.



One encrypted tunnel on every road,
two at once for 2 s during a switch

1 **Predict.** The coverage map shows the Wi-Fi edge; Infinet checks the next network 5 s ahead and switches with 1.5 s left.

2 **Check.** Red flags from the scan, then a probe in quarantine with a separate key. The verdict picks the road.

3 **Seal.** A WireGuard-style tunnel: tied to keys, not addresses, so the call survives every change of network.

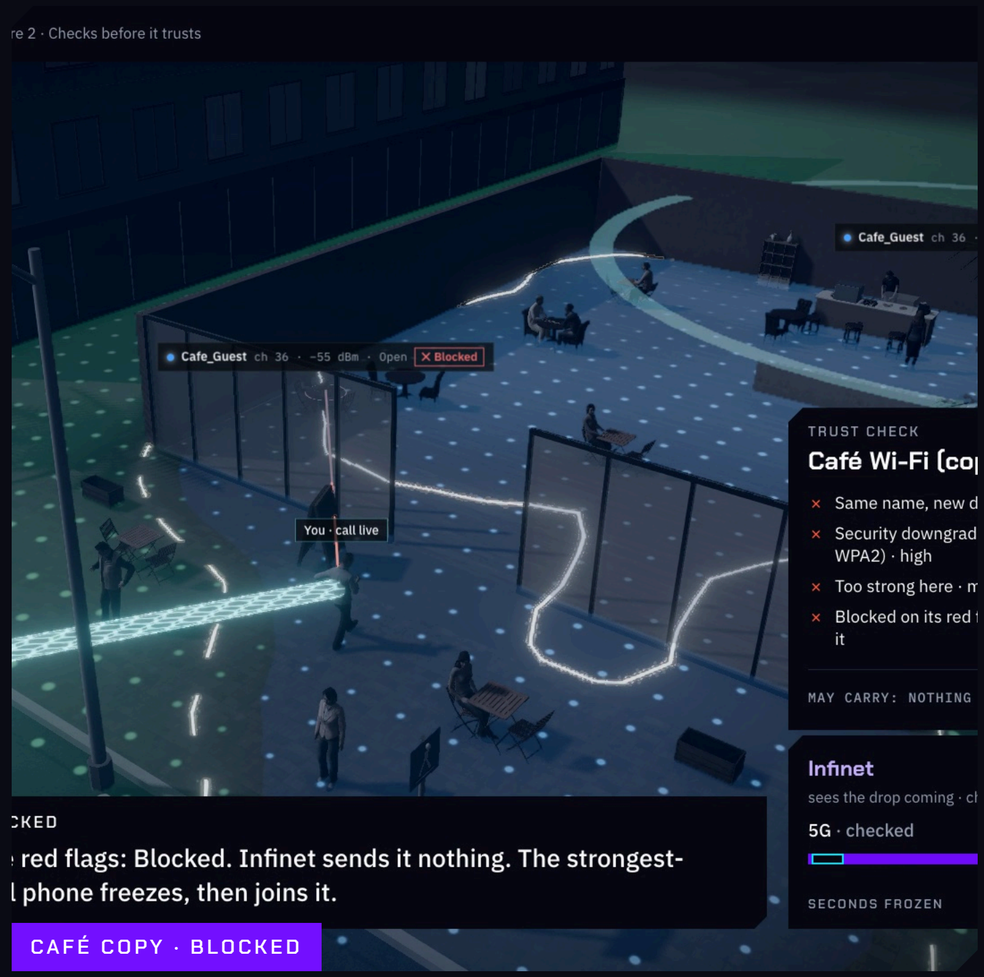
BEFORE A NETWORK CARRIES THE CALL

Five steps, and a fake network never gets the call.

1 · APPEARS	2 · SCREEN	3 · QUARANTINE	4 · VERDICT	5 · ROAM
Seen in a scan	Red flags, before anything is sent	Probe and canaries, separate key	One of five states	The call may move

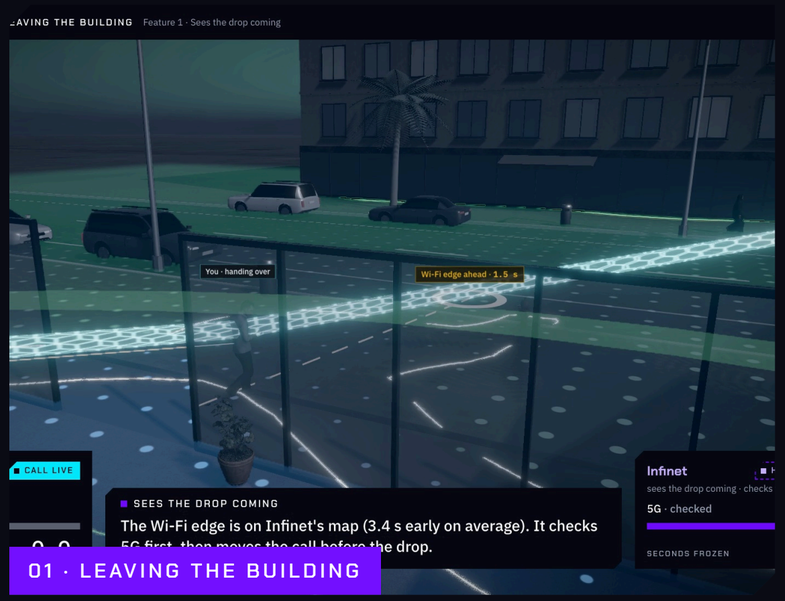
STATE	MEANING	MAY CARRY
CHECKING	Red flags and handshake running, or a sign-in page	Nothing yet
TRUSTED	Known network, handshake passed	The call
TUNNEL ONLY	Unknown network, handshake passed	Only our tunnel
BLOCKED	Handshake failed, or serious red flags	Nothing
NOT CHECKED	Other people's networks, seen, never joined	Nothing

Red flags: new device for a known name · security downgrade · weak encryption · signal too strong for the map · look-alike name · DNS mismatch. **One flag** caps at Tunnel only; **two serious** block before anything is sent.



A WALKABLE 3D WORLD · TWO PHONES ON THE SAME WALK

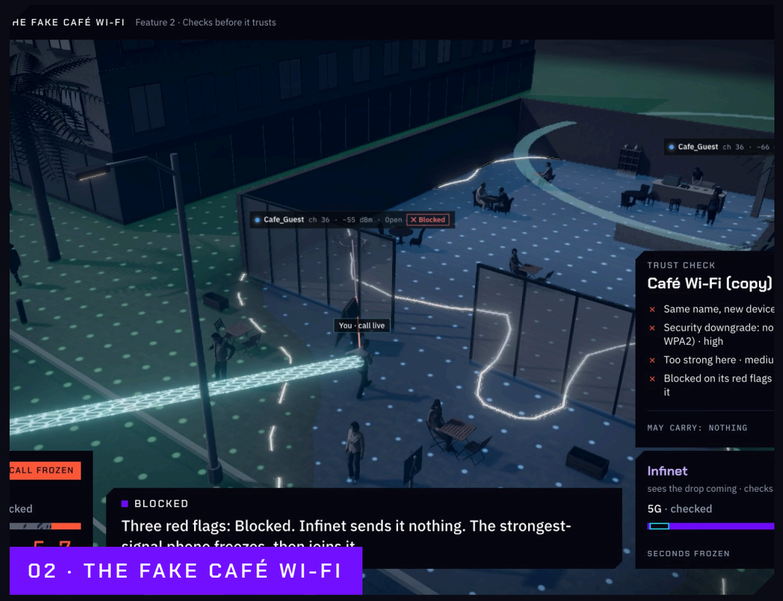
Three moments where today's phone fails, side by side with Infinet.



01 · LEAVING THE BUILDING

Sees the drop coming

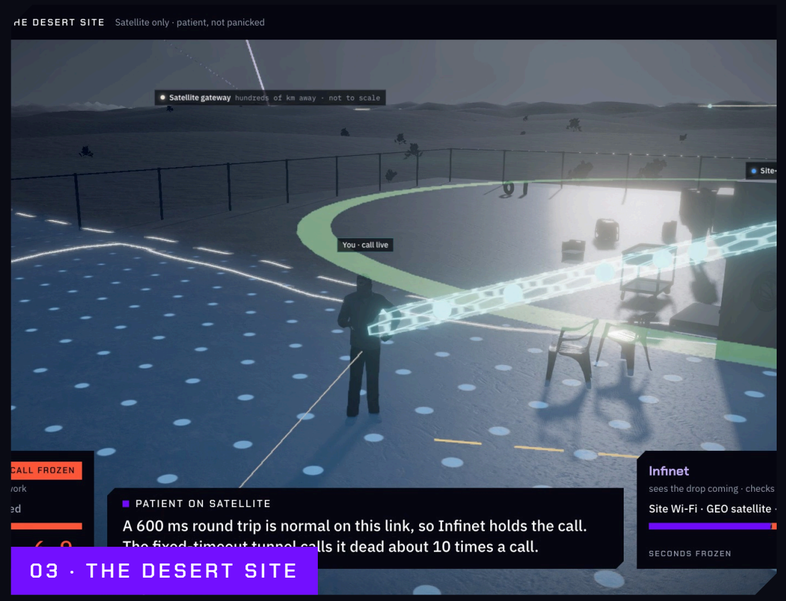
The Wi-Fi edge is on the map; Infinet checks 5G first and moves before the drop. **`0.0 s`** frozen.



02 · THE FAKE CAFÉ WI-FI

Checks before it trusts

Three red flags on the copy: Blocked, nothing sent. The strongest-signal phone joins it.



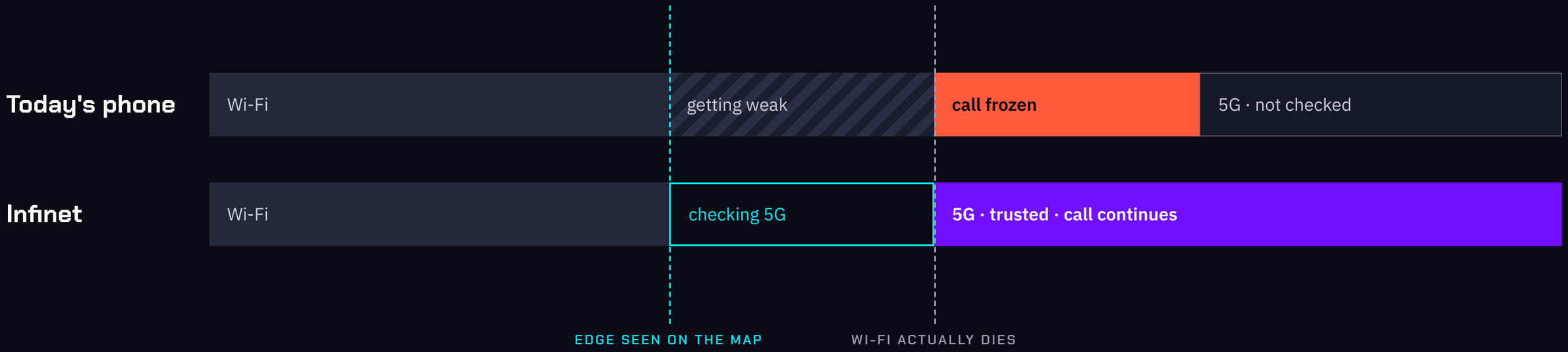
03 · THE DESERT SITE

Patient on satellite

A 600 ms round trip is normal here, so Infinet holds the call; a fixed timeout calls it dead.

SAME WALK, SAME SIGNAL · ILLUSTRATION, NOT MEASURED DATA

Today's phone waits for the Wi-Fi to die. Infinet moves before it does.



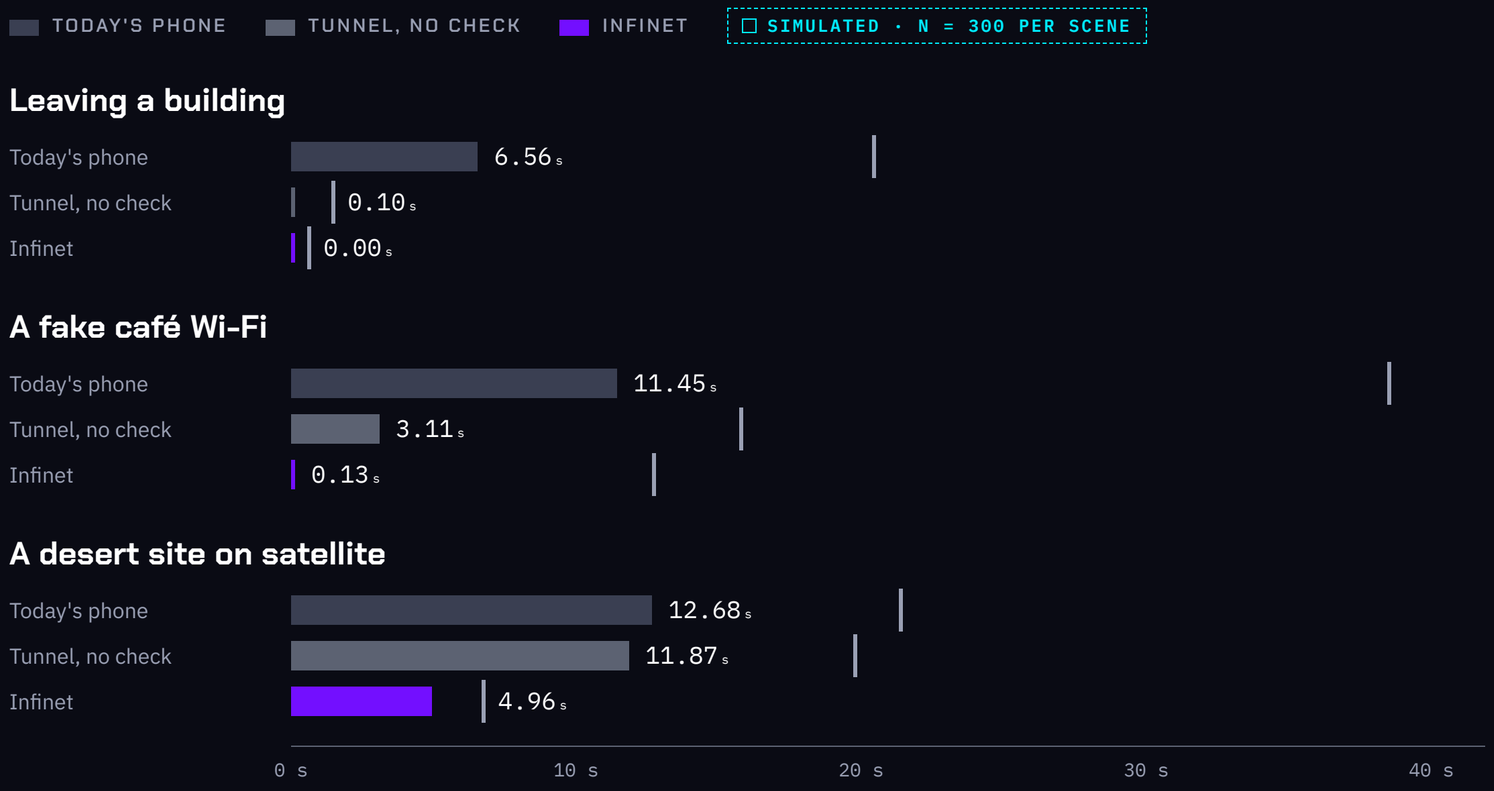
3.4 s
average warning before the Wi-Fi faded below −75 dBm

0.0 s
Infinet frozen leaving the building, against 6.56 s for today's phone

2 s
on both networks during the switch, so no voice is lost

SECONDS FROZEN PER CALL · BAR = AVERAGE · TICK = WORST OF 300 RUNS

Infinet cuts frozen time in every scene, and keeps every call.



CALLS SURVIVED · ALL THREE SCENES

900 / 900

today's phone: 427 / 900

VOICE FRAMES OVER A FAKE NETWORK

0

in 5,000 randomised runs, attacks included

THE COST OF BEING CAREFUL

+2–4.4 %

extra data from the 2 s duplication

Read honestly: one café run froze 12.7 s for Infinet; in the desert the call still freezes about 5 s during the satellite failover, but is never dropped.

HOW WE KNOW

A fair race: same walk, same radio, every phone.

FAIR COMPARISON

Common random numbers

Every policy rides the same walk, the same fading and the same attacker in each run. Only the decisions differ.

ENOUGH RUNS

300 runs per scene

Speed, start, shadowing, map error, attacker power and satellite timing all randomised. Mean, p95 and 95% bootstrap intervals.

SAFETY PROPERTY

5,000-run property test

Including relay attacks and forged disconnects: zero voice frames over a fake, unchecked or Blocked network.

ABLATIONS

Switch each part off

Without the trust check, Infinet accepts the fake café network in about half the runs (0.48). The check is what keeps it out.

GROUNDED INPUTS

Cited or labelled

3GPP TR 38.901 radio, ITU-T G.107 / G.114 voice, published LEO and GEO measurements. 96 of 163 settings are marked as assumptions.

REAL DATA

Our venue, measured

The hall's Wi-Fi is fitted to our own 27-point phone survey: 27 access points placed, 5 GHz within 3.8 dB.

WHAT EXISTS · WHAT WE ADAPTED

Phones and VPNs already move calls between networks. None checks a network before using it.

EXISTING MECHANISM	DOES WELL	MISSES	IN OUR RESULTS · SIMULATED
The phone's own switch Wi-Fi → mobile when Wi-Fi is poor	Built in, no setup	Moves after the call suffers; joins any saved name, even a copy	**6.56** s frozen leaving a building
Multipath TCP RFC 8684	Keeps a TCP connection over Wi-Fi and mobile	TCP only, not real-time voice; each app must support it	**1.6** s median reconnect, from iPhone measurements
Roaming VPN tunnel WireGuard, NDSS 2017	The call survives a new address, sealed	Doesn't know when to move, or which networks are safe	**8.29** switches per café call
Operator steering 3GPP ATSSS · TS 23.501	The operator's core splits traffic over 5G and Wi-Fi	Needs the operator's support; no help on café, venue or satellite networks	Not modelled
Enterprise roaming IEEE 802.11r	Moves between access points in under 50 ms	Only inside one network	Used inside Infinet

1 Check before use

Screen every new network, then probe it in quarantine with a separate key.

2 Move before it breaks

See the Wi-Fi edge on the coverage map; send on both networks for 2 s.

3 Seal everything

One tunnel on every network: even a perfect copy sees only ciphertext.

4 Patient on satellite

Adaptive timeouts (RFC 6298) and the learned LEO pattern, not a fixed timeout.

REFERENCES J. A. Donenfeld, “WireGuard: Next Generation Kernel Network Tunnel”, NDSS 2017 · 3GPP TS 23.501, ATSSS · IETF RFC 8684, Multipath TCP · IETF RFC 6298, retransmission timer

WHAT WE'D HAND AN OPERATOR OR A PHONE MAKER

Seven rules keep a moving call alive and private.

	RULE	VALUE WE USE	EVIDENCE ☐ SIMULATED · 300 RUNS PER SCENE
1	**Hold the voice target**	One-way delay ≤ 150 ms (ITU-T G.114); quality from the E-model (ITU-T G.107); conceal up to 2 lost frames	**4.34** vs 3.72 call quality (MOS) leaving a building · Infinet vs today's phone
2	**Prepare early, switch late**	Check the next network 5 s before the Wi-Fi edge; switch with under 1.5 s left; Wi-Fi usable above −75 dBm	**3.4 s early** average lead before the Wi-Fi faded · 0 s frozen
3	**Make before break**	Send on two tunnels for 2 s around a switch, only where the delay meets the playout deadline	**+2–4.4 %** extra data, by scene: the cost of the second copy
4	**Never use an unchecked network**	Screen, then probe in quarantine; timeout max(1 s, 3 × round trip); one flag caps at Tunnel only, two serious flags block	**0 frames** over a fake, unchecked or Blocked network in 5,000 runs
5	**Seal everything, fail closed**	Every frame in the tunnel on every network; anchor down → nothing in the clear, call ends after 8 s	**Ciphertext only** all that a perfect copy of the café Wi-Fi sees
6	**Be patient on satellite**	Adaptive timeouts (RFC 6298); learn the 15 s LEO pattern and protect 300 ms before each spike	**0** vs 10.4 false “link dead” calls per run · Infinet vs tunnel, no check
7	**Don't flap**	Wait 4 s before returning to a Wi-Fi, doubling to 60 s; Wi-Fi first only when checked and unflagged	**2.75** vs 8.29 switches per café call · Infinet vs tunnel, no check

Settings marked as assumptions are to be tuned with field data.

THE MAP INFINET NEEDS, BUILT FROM ONE PHONE WALK

We mapped this hall with one Android phone, and rebuilt it as a walkable twin.

1 · SURVEY
Our app listens only; a point every 4–5 m

2 · FIT
A radio model fitted to the scans

3 · REBUILD
The hall in 3D, with its crowd

4 · USE
Plan, rehearse, screen for fakes

27
survey points on set-up day, checked against a second phone

27
venue access points placed from the signal

3.8 dB
5 GHz fit to our measurements (RMS)

64
phone hotspots and 35 other devices, placed anonymously

~20
hotspots heard at any spot in the hall

4G
everywhere inside; never a 5G cell

A PRODUCT OF ITS OWN

Who would use it

- Venues and organisers, before a big event
- Security teams: every network on the floor screened
- Operators: measured indoor gaps, with the crowd
- Phones running Infinet: the coverage map itself

Other people's network names are never stored: each is a salted hash on the phone.

THE MARKET TODAY

Good tools exist, and every one is built for radio engineers.

PRODUCT	OWNER	WHAT IT DOES	LISTED PRICE
Ekahau	Ookla; Accenture agreed to buy Ookla (Mar 2026)	Wi-Fi design and survey, with its own survey hardware	≈ $1.5–2.5k a year + hardware ≈ $5k `reseller listings`
Hamina	Helsinki start-up, seed round 2025	Cloud Wi-Fi and private 5G planning, on-site survey	$590–980 a year per tool
NetSpot	Independent	Simple laptop Wi-Fi survey with heatmaps	≈ $149–499 one-off `reseller listings`
iBwave	Corning, since 2015	In-building cellular and Wi-Fi design; 600+ operators, integrators and vendors	On quote
Ranplan	Ranplan Wireless	In-building and outdoor planning; interactive 3D twins on NVIDIA Omniverse (MWC 2026)	On quote
Remcom	Remcom	Ray-traced radio propagation for indoor and urban twins	On quote
Rogue-AP detection	Network vendors	Finds rogue and “evil twin” access points, with sensors in the venue's network	Part of the network

Sources: vendor and press pages checked 10 Nov 2026; prices marked “reseller listings” were not confirmed with the vendor.

WHO IT'S BUILT FOR × WHAT IT SEES

Engineers are well served. Nobody sells venue owners the crowd's view.

↑ CROWD AND THREATS

MEASURE THE LIVE CROWD OR THREATS

Ookla Wind, RootMetrics | Rogue-AP detection

For operators and IT security, with their own sensors.

THE GAP WE FILL

Infinet Venue Twin

One phone walk, the crowd's hotspots, fake-network screening, and a 3D twin a non-engineer can walk.

ABU DHABI BUSINESS EVENTS, 2025

6,600

up 37%, with 2.2 million delegates (DCT Abu Dhabi)

CONSOLIDATION

Ekahau is part of Ookla, which Accenture agreed to buy in 2026; iBwave belongs to Corning.

DESIGN AND SURVEY TOOLS

Ekahau | Hamina | NetSpot | iBwave | Ranplan | Remcom

OWNERS · VENUE NETWORK ONLY

No product we found.

WATCH

Ranplan's 3D twins move towards owners. 3D alone is not our edge; the crowd and the trust check are.

RADIO ENGINEERS

VENUE AND EVENT OWNERS →

SIDE BY SIDE

The others each solve one piece. Infinet puts them together.

	TODAY'S PHONE	ROAMING VPN	MULTIPATH TCP	OPERATOR STEERING	SURVEY TOOLS	INFINET
Keeps the call across networks	◪	■	◪	■	—	■
Moves before the Wi-Fi drops	—	—	◪	◪	—	■
Checks a network before using it	—	—	—	◪	—	■
Seals every frame on every network	—	■	—	◪	—	■
Patient on satellite links	—	—	◪	—	—	■
Works on any café, venue or satellite network	■	■	◪	—	—	■
Maps a venue from an ordinary phone	—	—	—	—	◪	■
Shows the crowd's hotspots and fakes	—	—	—	—	◪	■

■ Yes ◪ Partly, or only in some setups — No

Our reading of each approach's public documentation; Infinet's column is simulated, not yet field-tested.

FROM SIMULATION TO A PILOT

Ninety days to a paying pilot, with one gate that decides.

WEEKS 1–2

Prove it

- Accuracy on survey points left out
- A real WireGuard handover on a phone
- Ten interviews with venues and organisers

◇ Gate: accuracy known, three want a pilot

WEEKS 3–8

Pilot

- Map one partner site or event
- Re-survey on the day; compare
- Test the price

◆ Gate: one paid pilot or letter of intent

WEEKS 9–13

Productise

- Automatic fitting from uploads
- A web viewer for the twin
- The Infinet app on real phones

◇ Gate: a second paying customer

BUILT	Decision logic, the simulation, the 3D world, the live scanner, the survey app
SIMULATED	The networks, the anchor server, the attacks
NEXT	A real handover on a phone, accuracy on unseen points, a pilot

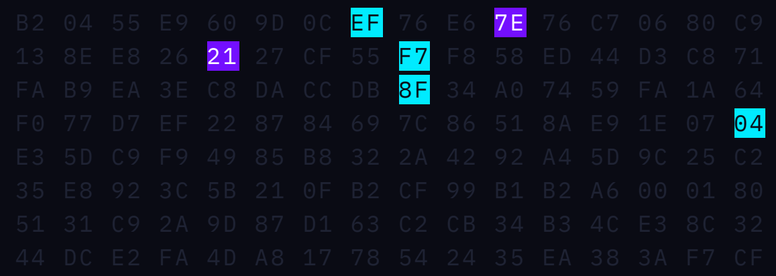
OUR ASK

One pilot: a building or an event of yours. We map it, deliver the twin and a report in two weeks, and you decide if it's worth paying for.

DECLARATION

We used AI, and we declare it.

Claude (Anthropic), through Claude Code, was our coding and writing assistant. The idea, the decisions and the survey are ours.



THE ASSISTANT

What it did

- Wrote most of the simulation code, tests and tools
- Built the pipeline that renders our film
- Drafted documents from our notes and decisions
- Most of our commits are marked as co-written with it

THE TEAM

What we did

- The idea and the problem framing
- The venue survey, walking the hall with our own phones
- The scenarios, the baselines and every design decision, logged with its reason
- Reviewed and tested what it wrote

HONESTY

How we checked it

- Every number traces to our settings file: cited, or marked as an assumption (**96** of **163**)
- **68** automated tests
- **300** seeded runs per scene, not hand-picked; demo runs chosen near the median

ASK US

Each of us can explain the tunnel, the trust check and the results in our own words, without the tool.

Stay connected.
Stay secure.

ابقَ متصلاً. ابقَ آمناً.

THANK YOU · QUESTIONS

Walk the demo with us at Stand 31: leaving the building, the fake café Wi-Fi, the desert site, and this hall.

TEAM INFINET · 42 ABU DHABI

DEVELOPED IN THE EDGE TRACK OF ATRC ATP 2026

